Qihui Feng

Curriculum Vitae

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Education

2022.04–Now **Doctoral Study, Computer Science**, *RWTH Aachen University*.
My study is funded by the Deutsche Forschungsgemeinschaft (DFG) under the supervision of Gerhard Lakemeyer. My research focuses on knowledge representation and reasoning in multi-agent systems. My research interests include:
- Multi-agent epistemic planning
- Modal reasoning and theorem proving
- Probabilistic reasoning
- Knowledge compilation
- Integration of LLMs with formal methods

2018.04–2021.03 **Master Computer Science**, *RWTH Aachen University*.
My project mainly focuses on: Knowledge Representation and Reasoning; Natural Language Processing; Automatic Speech Recognition. Master thesis: "Reasoning about Belief by Progression", which studies the progression reasoning in a noisy, dynamical environment.

2013.09–2017.06 **Bachelor Computing Science**, *Sun Yat-Sen University*.
A hybrid Study consists of 2 years of Mathematics and 2 years of Computer Science, including lectures of both majors like Programming, Data Structure, Algorithms, DSP, Computer Network, etc. Bachelor thesis: "Image Super-Resolution based on Dictionary Learning".

Teaching Experience

Teaching Assistant.
- Introduction to Artificial Intelligence (2022-2024)
- The Logic of Knowledge Bases (2023)
- Proseminar: Künstliche Intelligenz (2022, 2023, 2025)
- Seminar: Reasoning in Robotics (2024)

Thesis Advisor.
I have advised or am currently advising 5 Master's theses and 6 Bachelor's theses. One of the theses received the 2025 VCLA Outstanding Undergraduate Research Award.

Rewards

- KR2021 Marco Cadoli Distinguished Student Paper Prize

Job and Internship Experience

2021.03–2022.02 **Algorithmic Engineer**, *Midea Corp.*.
I was responsible for the development of a keyword spotting (KWS) model. The trained DFSMN-based model achieved a false reject rate of less than 5% (at 5m distance, 10 dB SNR, and normal reverberant environment) with zero false alarms during a 72-hour stress test. This system has been applied in Midea's ovens and other products such as refrigerators, homepods, and air conditioners.

2020.05–2020.08 **Intern**, *Midea AI Cloud*.
I contributed to the development of an industrial sound event detection (SED) system. Designed a novel SED framework based on neural Hidden Markov Model. The framework achieves good performance in both public dataset (DCASE2017, Challenge 3) and Midea's internal product anomaly detection task.

2018.12– 2020.03 **Student Assistant**, *Human Language Technology and Pattern Recognition Group, RWTH Aachen University*.
I designed a data filtering process for NER (Named Entity Recognition) data in the NIST TAC 2019 workshop and wrote the workshop paper as lead author.

Publications

- **Q. Feng***, G. Lakemeyer. Decidable Multi-agent Epistemic Planning: A Situation Calculus Approach [C]. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026.
- **Q. Feng***, H. Wilk, S. Khan, G. Lakemeyer. Translating Multi-Agent Modal Logics of Knowledge and Belief into Decidable First-Order Fragments [C]. *Proceedings of the International Conference on Autonomous Agents and Multiagent Systems 2025.*
- E. Jin*, **Q. Feng***, Y. Mou, S. Decker, G. Lakemeyer, O. Simons, J. Stegmaier. LogicAD: Explainable Anomaly Detection via VLM-based Text Feature Extraction [C]. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025, 39(4), 4129-4137.
- **Q. Feng***, G. Lakemeyer. Probabilistic Multi-agent Only-believing [C]. *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems.* 2024: 571-579.
- D. Liu*, **Q. Feng**. On the progression of belief [J]. *Artificial Intelligence*, 2023, 322: 103947.
- D. Liu*, **Q. Feng***, V. Belle, G. Lakemeyer. Concerning measures in a first-order logic with actions and meta-beliefs[C]. *Proceedings of the International Conference on Principles of Knowledge Representation and Reasoning.* 2023, 19(1): 451-460.
- **Q. Feng***, D. Liu*, V. Belle, G. Lakemeyer. A Logic of Only-Believing over Arbitrary Probability Distributions [C]. *Proceedings of the 2023 International Conference on Autonomous Agents and Multiagent Systems.* 2023: 355-363.
- D. Liu*, **Q. Feng**. On the Progression of Belief [C]. *Proceedings of the International Conference on Principles of Knowledge Representation and Reasoning.* 2021, 18(1): 465-474.(**Best student paper**)
- **Q. Feng**. Reasoning about Belief by Progression. Master's thesis, RWTH Aachen University, Aachen, 2021.
- **Q. Feng***, W. Wang, E. Tokarchuk, C. Dugast, H. Ney. Data Processing Techniques for the TAC-KBP 2019 EDL Task. *Text Analysis Conference (TAC)*, 2019.

*** Corresponding author(s)**.